



BRECHETEAU RESEARCH INSTITUTE

Chronon Field Program — Architecture, Claims, Tests, Reading Guide

The Chronon Field Program

Engineering contract for GR-subtracted, gauge-safe residual tests

Benjamin Br  cheteau

Orientation guide to the treatise *Time Is Not What We Think!* and the CHRONON-1 experimental pipeline
Contact: brecheteaub@gmail.com

Abstract — This document is a map, not the territory. It states the engineering contract for the Chronon Field program: a constrained, operational framework to test whether a residual tempo field exists once General Relativity and systematics are exhausted. Even if the hypothesis Φ is false, this pipeline provides a rigorous, reproducible audit for optical clock networks. It defines the admissible observables, the falsification gates, and the reading paths for different audiences. It is not a theory of everything; it is a theory of what comes next in precision metrology.

60-second summary (what this program is, now)

- **Claim (C0):** keep GR intact; introduce a tempo parameterization $\Phi(x)$ and a loss rate $\Gamma(x)$ used only in GR-subtracted, gauge-safe residuals.
- **First test (published):** CHRONON-1 Stage 00 loop-closure null tests.
- **Status (published):** Conservative integrity bound at the $\sim 10^{-17}$ level.
- **Why you should care:** reproducible integrity audit framework.

Glossary & units (minimum survival kit)

- $\Phi(x)$ — local tempo parameter, units s^{-1} .
- $\Gamma(x)$ — coherence/loss rate, units s^{-1} .
- τ — operational phase-time parameter, units s.
- u^μ — operational congruence.
- $X_\mu = \nabla_\mu \ln \Phi$, $\theta = u^\mu \nabla_\mu \ln \Phi$.
- L_{ABC} — loop-closure residual.
- Gates $G_0..G_3$.

Keywords — *Chronon Field, local tempo $\Phi(x)$, decoherence rate $\Gamma(x)$, operational metrology, optical clocks, time-transfer networks, null tests, reproducibility, reading map.*

A map before the territory

There are two ways to lose a reader in five minutes. You can give them a thick treatise with no entry point, or you can give them a clean pipeline with no conceptual frame. The Chronon Field Program contains both layers, by design. This document exists to connect them with a single promise: *you can understand what is being claimed, what is not being claimed, and what would falsify it, before you spend a week reading.*

I — What this document is (and what it refuses to be)

This document does four things, and refuses three.

It does:

- ✓ Fix the vocabulary and the non-negotiable invariants.
- ✓ Define what counts as an admissible observable.
- ✓ Show the experimental path (CHRONON-1 stages, gates, verdict logic).
- ✓ Give you a reading order depending on your background.

It refuses:

- × To sell an interpretation as if it were a result.
- × To hide uncertainty behind metaphors.
- × To use “new physics” as an excuse for loose metrology.

Target Audience:

- **Metrologists / Clock Teams:** Users who care about precise null tests, residual bounds, and network audits.
- **Quantum Decoherence / Optics:** Users interested in the phase/loss separation (Φ vs Γ).
- **Skeptics (GR-first):** Users who want a purely differential game where GR is the subtraction baseline.

Scope: this guide covers published artifacts and planned series. Only published artifacts carry citable DOIs; planned items are roadmap commitments, not evidence.

After Route 1 (15 minutes), you should be able to state: (i) the admissible observables, (ii) the first falsification gates, (iii) what would count as a failure.

If you are a lab, this document makes one decision easy: *is there a clean test here that you would be willing to run, audit, or try to break?*

If you are a lab: jump to Section X (Skeptic’s Menu) and pick a 2-day or 2-week test.

If you are a reviewer: use Route 2 (Section VII) as an attack surface.

II — The contract: invariants that keep the program honest

This section answers one question only: what are the non-negotiable invariants? The Chronon Field framework is deliberately constrained. These constraints are not decoration; they are the reason the program stays testable instead of dissolving into “theory-of-everything” fog.

(1) GR is intact. Geometry and null cones remain those of General Relativity. The framework does not introduce a modified metric, does not alter geodesics, and does not add an exotic stress–energy tensor.

(2) Phase and loss are separated. $\Phi(x)$ controls the tempo of phase evolution; $\Gamma(x)$ captures coherence loss. Confusing them destroys identifiability. Operationally, this is the split that keeps predictions sharp: clocks and interferometers read phase; coherence experiments read loss.

Think of Φ as “what your clock phase is doing after GR subtraction”, and Γ as “how fast coherence information is leaking out of your setup”.

Identifiability rule: any analysis where a re-parameterization can trade Φ against Γ without changing the observable is non-identifiable and cannot upgrade above C0.

(3) Multiplicative gauge. There is a global normalization freedom,

$$\Phi(x) \rightarrow \lambda \Phi(x), \quad \lambda > 0.$$

Physical content: only gauge-invariant ratios/gradients/loops of $\ln \Phi$.

(4) Differential-only reporting. The program does not compare “raw” data to imagination. It compares post-processed residuals to zero, after GR and known systematics are modeled and subtracted. If a claimed signal cannot be written as a residual, a ratio, a gradient, or a loop observable under a fixed operational gauge, it is not admissible inside this corpus.

Violation = immediate disqualification.

If you need to break any invariant to “explain” a result, it is not part of the Chronon corpus. Any claim that breaks these invariants is not “an extension”; it is a disqualification.

II.1 — Kill criteria (Immediate Falsification)

- **Robustness Failure:** If the signal disappears when you change the windowing or trimming parameters within reasonable bounds (robustness matrix) → **FAIL**.
- **Estimator Dependency:** If the result depends on a specific choice of estimator (e.g. median vs mean) → **FAIL**.
- **Ad Hoc Calibration:** If the signal requires a manual “calibration” term that cannot be derived from standard physics → **FAIL**.

III — Minimal backbone (the few equations you actually need)

This section answers one question only: what form do the primitives take? To keep the entry point clean, the backbone is intentionally short. Everything else in the corpus is an elaboration, a guardrail, or an application.

$$d\tau = \Phi^{-1}(x) dt, \quad J^\mu \equiv \Phi(x) u^\mu, \quad \nabla_\mu J^\mu = \Gamma(x). \quad (1)$$

All admissible observables downstream are built from these objects (derivatives of $\ln \Phi$ along u^μ and looped paths).

Dimensional Analysis (Standard SI):

$$[\Phi] = \text{s}^{-1}, \quad [\Gamma] = \text{s}^{-1}, \quad [\tau] = \text{s}.$$

Dimensional consistency checks kill 80% of invalid objections immediately.

What Φ and Γ are NOT:

Symbol	Is NOT...
Φ (Tempo)	Energy, Potential, Force, or “Time Flow” (it is a local clock rate).
Γ (Loss)	Friction, Heat, or “Arrow of Time” (it is strictly information loss).

Useful Invariant: If you see the ratio $\gamma(x) = \Gamma(x)/\Phi(x)$, this is the *dimensionless fractional loss per phase cycle*. It is a gauge-invariant diagnostic and useful for comparing loss across different frequency domains.

IV — Admissible observables: what can be measured without cheating

This section answers one question only: what you are allowed to plot and claim after GR subtraction. Most “new time” proposals fail for a simple reason: they do not specify what would count as a clean observable once the standard model of measurement has already spoken. Here the rule is strict: an observable is admissible only if (i) it survives the multiplicative gauge $\Phi \mapsto \lambda \Phi$, (ii) it is operationally defined on an instrumented window, and (iii) it is reported as a residual after full GR subtraction and known systematics.

The consequence is sharp: the corpus does not use $\Phi(x)$ directly. It uses gauge-invariant derivatives of $\ln \Phi$, transported along an operational congruence u^μ . The two primitives that recur everywhere are

$$X_\mu \equiv \nabla_\mu \ln \Phi, \quad \theta \equiv u^\mu \nabla_\mu \ln \Phi = \dot{\Phi}/\Phi,$$

and, when closed paths are instrumented, a clock-geometry one-form built from the same data. In practice, this means the Chronon Field shows up as (i) gradients, (ii) drifts, (iii) loop holonomies, never as an absolute “tempo meter reading”.

Why the gauge matters. If you normalize Φ arbitrarily, you can make it “fit” anything. Gauge discipline blocks that loophole. Only ratios, differences, gradients, or closed-loop effects remain. This is not philosophical; it is what prevents interpretive drift and keeps laboratory tests meaningful.

IV.1. The primitive base: three families (and why they are enough) Operationally, admissible observables fall into three families. They form a complete base on laboratory domains.

Observable	Data Required	What it cancels	Failure Mode
Loop (holonomy)	3+ clocks, pairwise links, closed path.	Shared noise, geometry path delays.	time-transfer modeling / estimator policy / geodesy inputs / thermal structure
Co-local ratio	2+ standards at same site.	Geometry, path delays.	servo/comb offset / environment coupling / uncontrolled local perturbations
Angular / Shape	Dimensionless geometric data.	Absolute scale dependence.	station coordinate drift / geometry model mismatch

Table 1 Admissible observables: the three operational families.

If your favorite candidate observable does not reduce to one of these three families under gauge and GR subtraction, it is not admissible.

IV.2. Forbidden observables (The “Do Not Fly” List) In practice: no single-clock drifts, no cosmic cross-correlations without local closure, no talk about “global tempo”. The following are strictly forbidden because they depend on gauge or absolute time:

- **Isolated Absolute Drift:** $\dot{\nu}/\nu$ on a single clock without a reference.
- **Cosmic Correlation without Loop Closure:** Correlating local noise with a cosmic signal before verifying local closure.
- **Global Tempo:** Arguing about the value of $\Phi(x)$ at one point (meaningless).

IV.3. Consequence: what the corpus will never claim at Stage 0 Stage 0 will never claim a detection (see FAQ Q5). It can claim only this: the network behaves as it should under null tests (closure, transitivity, robustness variants), and therefore it is worth attempting a constrained residual analysis at later stages.

V — CHRONON-1: the falsifiable pipeline (what labs can actually run)

This section answers one question only: what is the execution path? Stage 00 (already published) is the integrity gate: loop-closure null tests under multiplicity control and robustness variants. If the Chronon Field is the conceptual frame, CHRONON-1 is the execution contract. It is the part of the corpus that behaves like engineering: frozen inputs, pinned environments, explicit gates, and a verdict rule that does not depend on anyone’s enthusiasm. The priority is not “finding an effect”; the priority is establishing that the measurement network itself is consistent enough that a post-GR residual question even makes sense.

V.1. Why Stage 00 comes first (and why it is not optional) Any attempt to constrain a post-GR residual using distributed clocks or time-transfer links assumes a basic property: *transitivity*. If A compares to B and B compares to C, then A compared to C should be consistent. Stage 00 is a coherence audit. It answers one question only: *does the network close on itself under loop-closure null tests?*

V.2. The core observable of Stage 00 (loop closure) On a set of baselines $A-B$, $B-C$, $A-C$, define the loop observable:

$$L_{ABC}(d) \equiv \Delta_{A \rightarrow B}(d) + \Delta_{B \rightarrow C}(d) - \Delta_{A \rightarrow C}(d). \quad (2)$$

The null hypothesis is simple: $H_0 : L_{ABC} = 0$. Stage 00 does not interpret $L_{ABC} \neq 0$ as a discovery. It interprets it as a *failure mode to localize*.

Example: if $A-B$ and $B-C$ both close at the 10^{-17} level but $A-C$ does not, L_{ABC} tells you exactly how bad the inconsistency is.

V.3. Gates, robustness, and a verdict you can audit Two principles govern the verdict: (i) *Multiplicity control*: if you test many triangles, you pay the multiple-testing price. (ii) *Worst-case synthesis*: the reported bound is taken from the most conservative (worst) point in the robustness matrix.

Gates (pass/fail definitions — canonical)

- **G0 — Integrity (data + pipeline)**: The observable family is well-defined; environment+hashes match; closure/transitivity checks are computable and reproducible. *FAIL if: hash mismatch, missing metadata, non-reproducible outputs, or closure cannot be computed unambiguously.*
- **G1 — Identifiability (Φ vs Γ / model degeneracy)**: the reported residual bound does not hinge on a relabeling, hidden priors, or non-standard input models. *FAIL if: estimator/model choice changes the bound class or flips interpretation (degeneracy not resolved).*
- **G2 — Robustness (matrix)**: the result survives a pre-declared robustness matrix (windowing, trimming, aggregation, HAC, etc.) under worst-case synthesis. *FAIL if: signal/bound is driven by a small subset of variants or collapses under benign variants.*
- **G3 — Independent replication**: a second team reproduces the core artifacts from the pinned package (same hashes / regenerated figures / same verdict class). *FAIL if: replication cannot reproduce artifacts under the pinned environment.*

Verdict classes: PASS / FAIL / AUDIT / INCONCLUSIVE (see Section VIII.2).

V.4 — Stage map (what each stage is allowed to claim) The stages are not “chapters”. They are permission levels: each stage defines (i) allowed outputs, (ii) forbidden interpretations, and (iii) required gates.

Stage	Question	Admissible observable	Required gates	Allowed output	Forbidden
00	“Does the network close?”	Loop closure L_{ABC}	G0 (integrity)	PASS / FAIL / AUDIT + conservative integrity bound	Any “field detection” narrative
01	“Single-link endpoint residual?”	Endpoint-only residual (single link)	G0 + G1 (identifiability)	Residual bound + flagged systematics	Global inference, cross-link extrapolation
02	“Network-consistent constraint?”	Multi-link constraint + redundancy	G0 + G1 + G2 (robustness)	Global bound (worst-case across robustness matrix)	Claim upgrade if estimator-sensitive
03+	“Cross-platform convergence?”	Multiple families (loops/ratios/coherence)	G0..G3 (replication)	Only convergence-consistent constraints	Any upgrade without independent replication

Table 2 Operational rule: if a result changes materially under reasonable robustness variants, it stays Inconclusive (debug prompt, not evidence).

No stage below 02 can legitimately be sold as “detection” in this program.

V.5. Data Readiness Checklist (for Stage 01/02) Before attempting Stage 01 or 02, ensure:

- ✓ **Input Format:** Standardized CSV/Parquet with explicit timestamps.
- ✓ **Geodesy:** Station coordinates known to cm-level in ITRF.
- ✓ **Geometry:** At least one redundant baseline or closed loop.
- ✓ **Time-transfer model declared:** fiber/satellite policy, calibration, uncertainty budget.
- ✓ **Environmental logs available:** station and link thermal/pressure/humidity as applicable.
- ✓ **Estimator policy pre-registered:** windowing, trimming, multiplicity correction.

V.6. What makes CHRONON-1 attractive to a lab A lab does not need to agree with the Chronon Field interpretation to contribute. It only needs to accept a clean question and a reproducible artifact. CHRONON-1 is deliberately written so that the first engagement is adversarial by default: *try to break the null tests*. If you succeed, you improve the metrology. If you fail, you strengthen the integrity bound. Either way, the scientific value is immediate.

Concretely, there are three high-leverage contributions a lab can make without changing any philosophical commitment: replicate Stage 00 end-to-end from the pinned artifact; propose one additional null test that would be mandatory under your internal standards; and identify the first plausible hidden failure mode (geodesy inputs, environmental structure, time-transfer subtleties, or estimator policy) that could masquerade as non-closure.

The program is not asking for belief. It is asking for *good adversarial engagement*.

VI — The corpus architecture (how the pieces fit without confusion)

If you only want to know “what lives where” in the program, read Table 3 and then skip to the reading paths. The Chronon Field corpus is not a pile of papers. It is a program with a deliberate separation of layers.

Series	Mission	Inputs	Outputs	Ceiling	Gates
CHRONON-1	Falsifiable execution spine: integrity $\rightarrow$ residual $\rightarrow$ constraint $\rightarrow$ replication	Clock networks / time-transfer / geodesy (residuals)	Reproducible artifacts, verdicts (PASS/FAIL/AUDIT), bounds	C0 $\rightarrow$ C2	G0–G3
Chronon Field Series	Vocabulary + admissibility + anti-drift framing	Conceptual work attached to an observable family	Bridges (C0), non-claims, dictionaries	C0 $\rightarrow$ C1	G0–G1
Φ System Series	Hard formalism: axioms, symmetries, response, no-go, estimation	Formalism + instrument mapping (GR fixed)	Technical dossiers (axioms / dictionaries / estimation)	C1 $\rightarrow$ C2	G1–G2
Entanglement Dynamics	QM track under same discipline: decoherence / Born boundaries / non-Markov	Coherence data + defined admissibility	Testable signatures, protocols	C1 $\rightarrow$ C2	G1–G3

Table 3 The Program Series Map (structural; updated only if the architecture changes).

VI.1. The Series Map (Stable Structure) *Status policy: this table is structural; it is updated only when the program architecture changes. The full live catalog lives in the Program Index.*

In one line per series:

- **CHRONON-1:** what labs can actually run and replicate.
- **Chronon Field Series:** conceptual bridges, no claim upgrades alone.
- **Φ System Series:** hard formalism, theorems and estimation.
- **Entanglement Dynamics:** QM track under the same admissibility discipline.

VI.2 — Roadmap per series (cap, not list) This program is designed so that following it does not mean “reading everything”. It means tracking four parallel tracks with different deliverables.

CHRONON-1 (execution spine): expands from integrity (Stage 00) to constrained residual bounds (Stage 01/02) under strict gates. Expect releases that look like engineering: frozen inputs, pinned environments, explicit verdict logic, and replication-ready artifacts.

Chronon Field Series (conceptual bridges): continues to publish domain-mapping papers that (i) define vocabulary, (ii) prevent interpretive drift, and (iii) always point back to an admissible observable family. These papers do not upgrade claims by themselves.

Φ System Series (hard formalism): delivers the technical guardrails (axioms, no-go results, dictionaries, linear response, estimation/completeness) that make the program mathematically reviewable without ever touching GR geometry.

Entanglement Dynamics (QM track): extends the same admissibility grammar to decoherence / Born-rule boundaries / non-Markov structure, with an emphasis on “what would count as a clean observable” rather than narrative explanations.

Horizon	CHRONON-1	Chronon Field Series	Φ System Series	Entanglement Dynamics
2026	Stage 00 released; consolidate pipeline; package CHRONON-1 data; draft CHRONON-2 (results/data synthesis, Oct–Dec 2026)	Publish 2026 bridge arc (Feb–Oct 2026, up to “Year of Chronon”)	Prepare Φ System dossiers (outline + prerequisites; launch starts 2027)	Prepare QM track (admissibility grammar + protocol scaffolding; launch starts 2028)
2027	Release CHRONON-2 (CHRONON-1 data synthesis) + replication/engagement push	Consolidate corpus + clarifications	Launch Φ System series (Jan 2027–Jan 2028: Axioms $\rightarrow$ Paradoxes/Compatibilities $\rightarrow$ Completeness)	–
2028+	–	–	Complete Φ System (Completeness / Meta theorem, Dec 2027–Jan 2028)	Launch Entanglement Dynamics series (Mar 2028–Mar 2029: Born / non-Markov / coherence tests)

Table 4 Strategic Roadmap (cap, not exhaustive list).

VII — Reading paths (four routes, depending on who you are)

A reader does not need the same entry point as a lab. If you are a lab $\rightarrow$ Route 3. If you are a skeptical reviewer $\rightarrow$ Route 2. If you are a funding panel $\rightarrow$ Route 1 + Section V + Section X.

Route 1 — The “15 Minutes” Executive Scan (1) **1-minute abstract** (Page 1). (2) **The Contract** (Section II). (3) **Admissible Observables** (Section IV). (4) **The Stages & Verdicts** (Section V). (5) **Done**. You know what is claimed and how to break it.

Route 2 — The “Reviewer Hostile” Path (Attack Surface) If you want to find a flaw quickly: (1) **Kill Criteria** (Section II). (2) **Forbidden Observables** (Section IV.2). (3) **The Null Test Logic** (Section V.2). (4) **The Claim Ladder** (Section VIII).

Route 3 — Metrology / optical clocks (Practical) Start where scientific discomfort is productive: null tests. Read Stage 00 and the protocol logic first. Once you understand how the verdict is produced, you can decide if the residual question is worth your time.

Route 4 — Foundations / Philosophy Start with the invariants, not the poetry. This guide $\rightarrow$ The Contract $\rightarrow$ Treatise $\rightarrow$ Rhythm Leaves.

VIII — The claim ladder (C0 $\rightarrow$ C3) and what would count as falsification

This section answers one question only: how does a claim upgrade? A reader should never have to guess what is being claimed. The Chronon Field program is written as a ladder: each rung is a strictly stronger statement than the previous one.

Claim	Meaning	Minimum observable family	Gates	Typical stop condition
C0	“This is the grammar.”	N/A	G0	Any invariant violation
C1	“Upper bound on admissible residuals under null.”	Loop or co-local ratio	G0–G2	Estimator dependence / non-identifiable model
C2	“Structured residual signature.”	Loop + redundancy (≥ 1 closure + cross-check)	G0–G3	Non-replicable / platform-specific
C3	“Unified convergence.”	Multiple families across platforms	All	Conflicting platforms / missing closure

Table 5 Minimum artifact standard: pinned environment + multiplicity control + robustness matrix.

VIII.1 — Check matrix (minimum requirements per claim) Operationally:

- **C0:** you accept the grammar and invariants.
- **C1:** you accept an upper bound on residuals under null.
- **C2:** you accept a reproducible structured residual.
- **C3:** you accept a cross-platform convergence.

VIII.2. The “Inconclusive” Verdict It is vital to define failure modes that are neither “false” nor “true”. A result is **Inconclusive** if:

- It shows a signal but fails the robustness matrix (fails Gate G2).
- It shows a bound but relies on non-standard input models (fails Gate G1).
- It passes internally but cannot be replicated by a second team (fails Gate G3).

Inconclusive results must not be sold as evidence; they are debug prompts.

IX — Citations / versions / canonical references

IX.1. How to cite in 2 lines

Program: Brécheteau, B. (2026). *The Chronon Field Program* (v1.0). Zenodo. DOI: 10.5281/zenodo.18615897

Experiment: Brécheteau, B. (2026). *CHRONON-1 Stage 00 (Null Tests / Integrity)*. Zenodo. DOI: 10.5281/zenodo.18300382

IX.2 — Program Index (Living Catalog) The authoritative, up-to-date catalog of this program is the **Program Index** (public, pinned, citable Zenodo record). It is the only place where artifact status is declared and contains:

- A **Published** table: title, short label, version, date, DOI, and artifact type (paper / software / dataset / preregistration / SAP).
- A **Planned / Draft** table: items that may exist as manuscripts or internal drafts but are **not citable** until publicly released with a DOI.
- A **Redirect / Superseded** table: when a DOI is replaced by a newer record, the Program Index explicitly states what supersedes what.

Status definitions (strict):

- **Published** = public Zenodo record with a DOI (immutable artifact; safe to cite).
- **Planned / Draft** = not publicly released with a DOI (do not cite; do not treat as evidence).

Citation rule: cite only items explicitly marked **Published** in the Program Index. If an item is not listed as **Published**, treat it as non-existent for scientific purposes.

Program Index (public, pinned): 10.5281/zenodo.18615897.

IX.3 — Entry Set (The Stable Spine) The following items constitute the permanent core of the program (the “Spine”).

Conceptual Spine (The Argument)

- *Time Is Not What We Think! (Treatise, v2.0)*: 10.5281/zenodo.18378595

Experimental Spine (The Falsifiable Loop)

- *CHRONON-1 Stage 00 (Null Tests / Integrity)*: 10.5281/zenodo.18300382
- *CHRONON-1 Scientific Pipeline (Code)*: 10.5281/zenodo.18216849
- *CHRONON-1 Validation Plan*: 10.5281/zenodo.18146826
- *CHRONON-1 Statistical Analysis Plan*: 10.5281/zenodo.18072280
- *CHRONON-1 Experimental Pre-Registration (Restricted)*: 10.5281/zenodo.17571491

X — Laboratory engagement (a minimal test menu, designed for skeptics)

The Chronon Field program is deliberately built so that a laboratory can engage *without* endorsing the ontology. You do not need to believe in “tempo”. You only need to accept a clean operational question: once GR and known systematics are subtracted, do admissible residuals behave as strict nulls?

X.1. The Skeptic’s Menu (Engagement Levels)

- **Short (2 days)**: Replicate Stage 00 from the pinned artifact. Verify the hash. Regenerate the closure plots.
- **Medium (2 weeks)**: Design a co-located ratio test. Force a failure by heating one clock (injection test) to verify that the pipeline catches it.
- **Long (2 months)**: Run a full optical link campaign with redundant baselines to bound endpoint residuals under standardized geodesy.

Recommended first move for most labs: run the 2-day Stage 00 replication, then decide whether to invest in a 2-week injection test.

Replication deliverable: (i) environment hash match, (ii) regenerated figures, (iii) PASS/FAIL/AUDIT statement, (iv) one paragraph on first suspected failure mode, (v) minimum replication artifacts (3–4 livrables attendus).

X.2. Contribution Template (Standard Protocol) If you want to contribute, please format your request as follows:

Dataset: [e.g. 3-node optical fiber network, 14 days]

Observable: [e.g. Loop closure L_{ABC}]

Expected Failure Mode: [e.g. Temperature correlation]

Minimal Reproduction Steps: [Link to repo/data]

What would change your mind: [e.g. “If residual persists after thermal correction”]

Do not send open-ended philosophical questions. Send testable failure modes. If you send a contribution in this format, you will get a concrete, program-level answer rather than a philosophical discussion.

FAQ / Objections (technical, short, hostile-proof)

Q1 — Is Φ an energy potential or a new force? A: No. The program does not introduce Φ as a Hamiltonian potential nor as a stress-energy source. Φ is an operational clock-rate parameterization used only inside GR-subtracted, gauge-safe residual observables (ratios/gradients/loops). No energetic claim is made at Stage 00.

Q2 — Are you modifying General Relativity? A: No. Geometry and null cones remain GR. GR is the subtraction baseline. Any “explanation” requiring modified causal structure is outside the corpus.

Q3 — Why a multiplicative gauge $\Phi \rightarrow \lambda\Phi$? A: Because an absolute normalization of Φ is not observable. The physical content lives in gauge-invariant structures: derivatives of $\ln \Phi$, ratios, gradients, loop holonomies.

Q4 — Why endpoint/loop and not raw drift? A: Raw drift on a single clock is degenerate with reference frame, model choice, and local systematics. Loop closure and co-local ratios cancel entire classes of ambiguity by construction.

Q5 — What exactly does Stage 00 claim? A: Only network integrity: closure/transitivity under multiplicity control and robustness variants. Stage 00 never upgrades to “detection”.

Q6 — What is the first plausible hidden systematic? A: Time-transfer modeling (fiber delay calibration), environmental structure (thermal gradients), geodesy drift (cm-level in ITRF), or estimator policy. Stage 00 is designed to localize these failure modes.

Q7 — What makes a result “inconclusive”? A: Any signal that fails robustness (G2), identifiability (G1), or replication (G3) is inconclusive and must be treated as a debug prompt, not evidence.

Q8 — What would falsify the program (hard stop)? A: (i) Any required violation of the invariants (must modify GR or break gauge discipline to explain residuals), or (ii) repeated, independent null constraints that push admissible residual bounds so low that no non-trivial operational content remains (program becomes metrology audit only).

Q9 — Does a tight null bound imply $\Phi = 0$? A: No. A null bound implies “not detectable in this observable class at this sensitivity”. It constrains operational content; it does not prove an ontological zero.

Q10 — Why keep conceptual series (Chronon Field Series) if they don’t prove anything? A: They map domains and identify which admissible observable family could carry a test. They are C0 only unless explicitly tied to a CHRONON stage + pinned artifact.

Q11 — Where should a reviewer attack first? A: Identifiability (can Φ vs Γ be re-labeled?), estimator dependence, multiplicity control, and time-transfer systematics.

Q12 — What is the minimum valuable contribution from a skeptic? A: A replicated Stage 00 run + one additional mandatory null test + a concrete candidate failure mode.

Q13 — Why not assume it’s all time-transfer systematics? A: That is the default hypothesis. The entire program is built to force that explanation to fail before any upgrade: loop closure, redundancy, robustness matrix, multiplicity control, and replication gates are designed to expose transfer-model artifacts first.

Q14 — If Stage 00 is “only integrity”, why publish it? A: Because integrity is the prerequisite for any residual question. Stage 00 is the “license to ask” later-stage questions; without it, residual claims are scientifically meaningless.

